

Lab 4b - Analysis of people's fears

The data for this lab is part of a large dataset collected by researchers at Chapman University based on what people fear. Because of this, it's got a lot of interesting questions to explore. A large sample like this can give a good idea about the thoughts of a population, which in this case is American adults in 2022.

More relevant to psychology would be some of the relationships between the variables and looking at the categories. In this lab, we'll discuss how we might look at some of these categories and how R allows people to investigate categorical and ordinal data.

In this lab, there are a few packages that are a bit different than what we're used to.

Step 1: Go to the right side of the screen and click on the packages tab. Search for and install the "haven" package.

Step 2: Create a code chunk which loads the tidyverse library and the haven library, like what is listed below

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.2      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(haven)
```

Step 3: We are going to load the data as follows. Create a code chunk like below using the following code. Note in this case, the data format is a bit different, and that's why we are using a different code chunk. The original data isn't in CSV format but this will allow you to see what the variables are called.

```
d = read_spss("lab4b.sav")
```

A guide to the variables in this dataset is here <https://thearda.com/data-archive?fid=CSAF2022&tab=2>. This gives us a useful guide to what the variables are called and their own frequency data. In this case, we are going to examine a few variables listed below:

- I_GENDER - gender, coded 1 male and 2 female
- I_MARITAL - marital status, coded 1 Single, 2 single living with partner, 3 married, 4 separated/divorced/widowed
- Q1 - how religious does a person consider themselves to be, 1 - not at all religious, 2 not too religious, 3 somewhat religious, 4 very religious

And there are many variables about fear. Here are a few that I'll use for the analysis below but you can find others on the page linked above. Each of these are scored 1 - very afraid, 2 - afraid, 3 - slightly afraid, 4 - not afraid at all (note higher numbers are less fearful)

- Q10C - fear of dying
- Q13A - not having enough money for the future
- Q14B - technology I do not understand
- Q15B - a devastating hurricane (Q15C is tornado)
- Q16A - fear of blood
- Q16C - insects, spiders, and arachnids
- Q16D - reptiles, such as snakes
- Q16F - sharks
- Q16G - germs
- Q16J - ghosts
- Q16N - zombies

In this lab, I want to walk through a couple analyses with a variable and then you can walk through the same analyses picking a different variable of interest. We will start with the variable **Q16K** or fear of heights.

The first way to look at categorical or ordinal data is to use the `table()` function to see how frequent a value is. Type the following to see what percent of people had each response for their fear of heights. Remember that 1 is very afraid.

```
table(d$Q16K)
```

```
##
##    1    2    3    4
## 146 168 372 334
```

It can be difficult to see patterns in the raw numbers, so we can use the `prop.table` function to see what proportion had each response. In this case, I'm doing something called **nesting** where I put one function inside of another function. R starts by evaluating the inner function and then the outer one.

```
prop.table(table(d$Q16K))
```

```
##
##           1           2           3           4
## 0.1431373 0.1647059 0.3647059 0.3274510
```

Okay, now let's look to see if fears of heights vary based on gender. I want to create a table for each gender and see if there's differences in the patterns. We can give two variables to table and get a frequency table broken down by the number of people in each of two cases.

```
table(d$Q16K, d$I_GENDER)
```

```
##
##           1    2
##    1  57  88
##    2  71  95
##    3 178 190
##    4 167 159
```

This tells us the number of people in each cell, that is men who are very afraid of heights (first row, first column). We might want to look at the frequency for each cell. In this case a basic `prop.table` might not work well:

```
prop.table(table(d$Q16K, d$I_GENDER))
```

```
##
##           1           2
##  1 0.05671642 0.08756219
##  2 0.07064677 0.09452736
##  3 0.17711443 0.18905473
##  4 0.16616915 0.15820896
```

Here we want to look at the percent of females who are in each category (very afraid, somewhat afraid, etc.) We can do that by adding an option to `prop.table()`. We type `margin = 2` because we want to break it up by gender which is the second variable.

```
prop.table(table(d$Q16K, d$I_GENDER), margin = 2)
```

```
##
##           1           2
##  1 0.1205074 0.1654135
##  2 0.1501057 0.1785714
##  3 0.3763214 0.3571429
##  4 0.3530655 0.2988722
```

Based on this, it seems like women are more likely to say they're afraid of heights than men. We can test if this difference is significant using a chi-square test. Remember, a chi-square test only indicates whether the two variables are dependent, but not the direction of the effect. We do this on frequency table in the `table` command

```
chisq.test(table(d$Q16K, d$I_GENDER))
```

```
##
## Pearson's Chi-squared test
##
## data:  table(d$Q16K, d$I_GENDER)
## X-squared = 7.2464, df = 3, p-value = 0.06445
```

In this case, we can see the p-value is not below .05 so we would conclude that this difference is not statistically significant, or not unlikely to be by chance alone.

But this is a great example of a *marginally significant* finding which is a finding that is almost significant. It doesn't meet the criteria of significance so we fail to reject the null hypothesis, But since it's close to being significant, it's somewhat likely this is due to low power. In this case, there's a huge sample size, so power is probably not an issue.

For the rest of the lab, I want you to pick two other variables and see what percent of people say they're very afraid, afraid, somewhat afraid, or not afraid of the variable and then see if it varies based on gender, marital status, or religion. For example, are religious people more afraid of zombies. Make sure you do the following:

1. List your two variables as annotations and give me an idea of what you predict, and why. I'm not judging whether you're making good predictions and you might not have a clear prediction.
2. Create a table of the number of people with each level of the fear. Also create a proportional table using `prop.table()` for just the fear
3. Use table to list the number of people in each level of fear for each group of your condition (e.g. number of married people who say they are afraid of ghosts).
4. Create a proportional table with the numbers in step 3, similar to above. Make sure it has the correct margin.
5. Conduct a chi-square analysis to see whether the two variables are independent or dependent of each other and report whether the effect is statistically significant, using $\alpha = .05$.
6. Knit this and submit the knitted document!